Tribhuvan University
Faculty of Management
M. Phil. in Public Administration, 2082

First Semester Examination — Full Marks: 60
MPH 502: Research Methods in Public Administration-II — Time: 4 hours

Read the instruction and answer accordingly
This exam paper is divided into two parts. Part A is theory-based and Part B computer based.
Unless and otherwise stated, all notations/ terminologies have their usual meanings
Answer sheet of both part must be submitted

Part "A"

[Full Marks: 30] **[time: two hours]**

Attempt any five questions

1. A local government office in Pokhara Metropolitan City receives citizen complaints about various municipal services. The response time (in hours) for addressing these complaints follows a normal distribution with a mean of 24 hours and a standard deviation of 6 hours.

   a) What percentage of complaints are resolved within 18 hours? b) What is the probability that a randomly selected complaint takes between 20 and 30 hours to resolve? c) The mayor wants to set a service standard such that 95% of complaints are resolved within a specific timeframe. What should this timeframe be?

2. The Department of Traffic Police wants to test if a new traffic management system has reduced average waiting time at major intersections. Before implementation, the average waiting time was 4.5 minutes. After implementation, a sample of 25 measurements showed: Sample mean = 3.8 minutes, Sample standard deviation = 1.2 minutes

   a) State appropriate null and alternative hypotheses. b) Conduct the test at $\alpha = 0.05$ and interpret the results. c) What assumptions are required for this test to be valid?

3. A government training institute is evaluating four different teaching methods for civil service training. The final examination scores of participants are:

Method	**A**	**B**	**C**	**D**
Mean Score	85.2	78.6	82.4	79.8
Sample Size	20	18	22	19
Std. Dev.	8.5	7.2	9.1	8.0

   The ANOVA table shows:
   - SS(Between) = 542.8
   - SS(Within) = 1,247.3
   - Total observations = 79

   a) Write the ANOVA table. b) Test if there are significant differences among teaching methods at $\alpha = 0.01$.

4. The Ministry of Finance is studying the relationship between government expenditure on education (in billions NPR) and literacy rate (%) across 77 districts of Nepal.

   Simple Regression Results:

   - Regression equation: Literacy Rate = 52.3 + 2.8 × Education Expenditure
   - $R^2 = 0.67$, Standard error of regression = 8.4
   - Standard error of slope = 0.45
   - Standard error of intercept = 3.2

   a) Interpret the slope coefficient in practical terms. b) Test whether education expenditure has a significant effect on literacy rate at $\alpha = 0.05$. c) Construct a 95% confidence interval for the slope coefficient. d) Predict the literacy rate for a district with education expenditure of 15 billion NPR. e) Calculate and interpret the coefficient of determination.

5. The Election Commission of Nepal conducted a post-election survey to analyze voting patterns across different demographic groups. The following data was collected from 500 randomly selected voters:

   Voting Pattern by Age Group:

Age Group	Party A	Party B	Party C	Total
18-30	85	45	20	150
31-50	70	60	45	175
51+	45	55	75	175
Total	200	160	140	500

   a) Find out the expected frequencies under independence? b) Test whether voting preference is independent of age group at $\alpha = 0.05$. c) Discuss the political implications of your findings.

6. The Local Government and Social Development Office wants to assess whether citizen satisfaction with grievance redressal differs between urban municipalities and rural municipalities. The satisfaction scores (1–10) from a random sample of citizens are:
   Urban: 8, 7, 9, 6, 7, 8, 6
   Rural: 5, 4, 6, 5, 4, 3, 5
   a) State the null and alternative hypotheses.
   b) Conduct a Mann–Whitney U Test at 5% significance level and Interpret the result.
   c) Why might the Mann–Whitney U test be chosen over a t-test in this scenario?
7. How can public administrators use statistical tools to make sense of survey data collected from a sample of citizens?

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Part "B"

[Full Marks: 30] **[time: two hours]**

Attempt all questions

You have given a cleaned dataset of a study on the effectiveness of a cognitive training program conducted in a large organization. The dataset contains 300 records of employees from different departments. The key variables are:

1. **Age** (Quantitative)
2. **Gender** (Categorical: Male, Female)
3. **Department** (Categorical: HR, Finance, IT, Operations, Marketing)
4. **Training_score** (Numeric: score from 5–20)
5. **Memory_test_before** (Numeric: score before training, 5–25)
6. **Memory_test_after** (Numeric: score after training, 5–25)

You are provided with the dataset in two formats: cognitive.sav and cognitive.csv.
Perform any statistical software and give the answer to the following questions.

A: Descriptive Statistics (4 Marks)

1. Create frequency and percentage tables for the categorical variables Gender and Department. (2)

Variable	Categories	Frequency	Percentage (%)
Gender	Male		
	Female		
Department	HR		
	Finance		
	IT		
	Operation		
	Marketing		
Total			

2. Find the maximum, minimum, mean, and standard deviation of the quantitative variables age, Training Score, Memory Test Before and Memory Test After. (2)

Variable	Minimum	Maximum	Mean	Standard deviation
Age				
Training Score				
Memory Test Before				
Memory Test After				

B: Categorical Analysis (6 Marks)

3. Categorize the age into 20–30, 30–40, 40–50, 50+ with upper limit excluded form and obtain the frequencies and percentage of this distribution. (3)

Age Group	Number of Persons	Percentage
20-30		
30-40		
40-50		
50+		
Total		

4. Obtain the bivariate distribution of Department and Age Group, and test whether there is an association between these two variables using chi square test. (3)

	Age group					
Department	HR	Finance	IT	Operation	Marketing	Total
20-30						
30-40						
40-50						
50+						
Total						
Chi Square		Conclusion:				

C: Correlation & Comparisons (10 Marks)

5. Find bivariate pearson correlation coefficients among the variables **`training_score`**, **`memory_test_before`**, **`memory_test_after`** (2)

Pearson Correlation Coefficients

	training_score	memory_test_before	memory_test_after
training_score			
memory_test_before			
memory_test_after			

6. Find average training score of male and female along with standard deviation also test whether there is significance difference in average value? Use Levene's test or F-test to check homogeneity of variance before applying t-test. (5)

Category	Average Score	SD Score	F-value	P-value
Male				
Female				
Discussion:				

Significance of difference of means

Measure	Average difference	t-value	p-value	95% CI Lower	95% CI Upper
Difference of means					
Discussion:					

7. Using a paired samples t-test, check whether there is a significant improvement in memory test scores after training. Report the average, SD, t-value, p-value, and interpret. (3)

Measure	Average	SD	t-value	P-value
memory_test_before				
memory_test_after				
Discussion:				

D: ANOVA and Regression (10 Marks)

8. Test whether there is significant difference in training score according to departments?(3)

Departments	Mean	SD	F-value	P-value
HR				
Finance				
IT				
Operation				
Marketing				
Discussion				

9. Conduct a multiple regression analysis using memory_test_after as the dependent variable and training_score, memory_test_before, and age as independent variables. Obtain R^2, Adjusted R^2, and standard error of estimate. Discuss the values obtained. Obtain the ANOVA table for this regression model. Discuss the overall significance of the model. Fill in the coefficient table with t-values and p-values. Describe the meaning of the regression coefficients obtained. (7)

R^2 and SE Table

Parameter	Value	Discussion
R^2		
Adjusted R^2		
Standard Error		

ANOVA Table

ANOVA	DF	Sum of Square	Mean SS	F value	P value
Regression					
Residual					
Discussion:					

Coefficient Table

Variable	Coefficients	Standard Error	t-value	P-value
Intercept				
training_score				
memory_test_before				
age				
Discussion:				